

Digital Twin for Synthetic Training Data Generation in C-SMC Strip Deposition: A Physics-Based Simulation Approach

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Abstract

The development of inline quality assurance systems for Sheet Molding Compound (SMC) manufacturing is hindered by the lack of large-scale, accurately labeled datasets required for training machine learning models. This work presents a physics-based digital twin framework for generating synthetic training data of carbon-fiber SMC strip deposition processes. The simulation framework models the complete deposition process, including gravitational settling, friction, belt motion, and controllable lighting conditions, while generating fully annotated images with multiple synchronized annotation formats: YOLO bounding boxes, point annotations, density maps, and polarization-based renderings (Angle and Degree of Linear Polarization). The framework's capability is demonstrated through the generation of over 5,000 rendered images with comprehensive ground truth annotations. Validation results confirm that the synthetic datasets replicate key industrial

characteristics, including quasi-isotropic fiber orientation distributions consistent with commercial SMC sheets (χ^2 test, $p < 0.001$) and realistic temporal density evolution profiles. Preliminary transfer learning experiments using YOLOv11 for object detection and CSRNet for density estimation demonstrate that models trained exclusively on synthetic data can successfully transfer to real manufacturing scenarios, despite an identified domain gap in intensity distributions. The automated generation pipeline enables rapid iterative refinement based on real-world feedback, substantially reducing the labeling burden compared to manual annotation approaches while providing a robust foundation for developing camera-based inline inspection systems for fiber orientation and fiber volume content estimation in SMC production.

Keywords: Sheet Molding Compound, Digital Twin, Synthetic Training Data, Physics-based Simulation, Machine Learning, Quality Assurance, Fiber Orientation Distribution, Polarization Imaging, Deep Learning, Computer Vision, Carbon Fiber Composites, Inline Inspection

1. Introduction

For decades, the global emphasis on reducing carbon dioxide emissions and improving energy efficiency has driven the transportation sector to shift toward lightweight construction. This long-term effort has established fiber-reinforced polymers (FRPs) as key alternatives to traditional materials like metals. Among FRPs, Sheet Molding Compound (SMC) parts are particularly attractive because they offer a balance of high specific stiffness and strength, low cost, and efficient manufacturability. SMC is a prepreg sheet: a semi-finished fiber-reinforced thermoset polymer matrix product that typically consists of a thermosetting resin (such as unsaturated polyester, vinyl ester, or epoxy), particulate fillers (e.g., calcium carbonate), and long, discontinuous fiber reinforcement (typically 25–50 mm in length).

The production of an SMC part begins with the creation of the semi-finished prepreg sheet on a compounding line, where the thermoset resin paste, fillers, and long chopped fibers or fiber bundles are combined and compacted onto carrier foils. The sheet then undergoes a maturing phase, which increases its viscosity to the level required for subsequent handling. After maturation, the SMC sheet is cut to the required size and, if necessary, preformed before being compression-molded into the final component.

The mechanical performance of an SMC component—particularly its strength and stiffness—is governed directly by the microstructure of the semi-finished sheet. This microstructure is described by two spatially varying fields: the fiber orientation distribution (FOD) and the fiber volume content (FVC). The FOD is a tensor-valued field, typically represented by a second- or higher-order orientation tensor that characterizes the local directional probability distribution of fibers or fiber bundles [1, 2]. The FVC, in contrast, is a scalar field specifying the local fiber volume fraction within the composite [3]. Together, these fields determine the local anisotropic mechanical behavior through established micromechanical homogenization relations [4, 5] and serve as the primary predictors of stiffness, strength, and failure-relevant properties at the component scale. However, local deviations from the expected orientation distribution or from the nominal fiber volume content inevitably arise due to the stochastic nature of fiber placement during sheet assembly. Additionally, it is still subject to debate whether a stochastically perfect process would yield a planar quasi-isotropic FOD [6, 7] or whether the forward feed in the direction of production would result in a non-isotropic FOD [8, 9].

Still, quality assurance during SMC manufacturing needs to define boundaries for possible deviations from the expected FOD and FVC and to provide efficient and precise means for measuring both. Therefore, in-line sensing of FOD and FVC during SMC sheet production is of great interest for quality assurance. Such an inline system must, however, satisfy several practical constraints. It must operate at typical line speeds of 1–10 cm/s, implying real-time data acquisition with minimal processing latency. It also requires sufficient spatial and angular resolution to detect deviations in fiber architecture, while remaining robust to surface variations, occlusions, and the inherent heterogeneity of discontinuous-fiber materials. A wide range of non-destructive testing (NDT) methods has been proposed, each with different spatial/temporal resolution, accuracy, and practicality [10, 11, 12, 13, 14, 15, 16].

While radiographic or tomographic methods can access bulk information [10], optical inspection approaches have emerged as attractive alternatives due to their non-contact nature and high acquisition rates [8, 16, 17, 18]. However, conventional optical approaches often require an extensive post-processing [19], which limits their suitability for inline use. Polarization imaging therefore represents a particularly promising alternative to conventional optical methods for carbon-fiber-reinforced materials, as carbon fibers

induce a pronounced polarization state in reflected light [20]. This enables fiber-orientation information to be extracted with substantially reduced post-processing effort, making the approach inherently better suited for inline applications [21].

Machine-vision cameras with on-chip polarizers can image the angle of linear polarization (AoLP) of light from carbon fibers, providing pixel-level surface fiber orientations. These systems are inexpensive (1–3 kEUR) and operate at video rates, enabling real-time inspection. For uni- and bidirectional C-SMC laminates, polarization-based orientation measurements have been validated with an angular accuracy on the order of 1° [18, 22]. Surface and bulk fiber orientations often correlate in C-SMC, but the method does not work for glass-fiber SMC, as glass fibers do not polarize light significantly. Nonetheless there are approaches to capture the drop-down area to gain more depth information of the SMC-Prepreg [23]. Furthermore, the method does not perform explicit object recognition and therefore remains sensitive to occlusions, surface curvature and local texture variations, which can bias the AoLP signal and cause certain regions to dominate the orientation estimate. While highly effective for surfaces with clear carbon-fiber texture, its robustness decreases for multidirectional or irregular SMC materials, where overlapping fiber families and varying reflection properties can lead to averaged or unstable orientation fields [8]. Post-processing of the polarization images can mitigate these effects, but such steps reduce the method’s potential for inline application due to increased computational effort and latency. Pre-trained machine-learning models offer a possible alternative, yet they require large amounts of accurately labeled ground-truth data [24] — data that are difficult or impossible to obtain experimentally during inline SMC production. Synthetic microstructure models therefore appear as a suitable source of labeled data.

Digital models for SMC prepregs have been proposed, typically generating microstructures by distributing fiber chips (pseudo-)randomly according to prescribed density functions [25, 26]. While suitable for statistical analyses, such models do not capture process- or appearance-related variability such as surface topology and local thickness variations, illumination effects, and deposition dynamics. Such factors, however, are known to be essential for training machine-learning models in image-based inspection tasks, as ML methods require realistic variations in texture and lighting to generalize effectively [27]. To address these limitations, the present work introduces a simulation framework in which fiber chips are not placed directly but are

physically dropped onto a moving conveyor belt, incorporating effects such as gravitational settling, friction, belt motion, and controllable lighting conditions. This enables the generation of realistic, fully annotated synthetic image datasets for supervised learning of fiber-orientation and fiber-volume-content estimation. The simulation framework can also be used to generate annotated images for conventional optical inspection approaches, thereby enabling benchmark datasets for both polarization-based and purely optical fiber-orientation analysis. The resulting digital representation of the SMC semi-finished material can be integrated into a virtual process chain—such as press forming simulations—to assess how the initial fiber architecture affects press parameters and the material properties of the final component.

2. Research Questions and Approach

Analyzing the current state of the art, the major obstacles in developing a real-time optical quality inspection system for SMC production are the lack of large-scale, accurately labeled datasets, the variability of fiber microstructure in multidirectional SMC sheets, and the need for computationally efficient inline methods. Therefore, this paper addresses the following research questions:

1. How can large-scale, fully annotated image datasets of SMC semi-finished parts be generated *in silico*, capturing realistic microstructural variability, lighting, and surface effects?
2. How can these datasets support the development and evaluation of camera-based, real-time quality assurance methods for SMC production?
3. How can machine learning algorithms for fiber-orientation and fiber-volume-content estimation be trained and validated using such synthetic data?

The research approach pursued in this work is the development of a scientific artefact in the form of a *digital twin* of the SMC manufacturing process. Classical computational approaches for fiber-orientation estimation are often too slow for inline application; by contrast, machine learning-based inference can achieve real-time performance once trained. The digital twin models the deposition and arrangement of fiber chips on a moving conveyor, including gravitational settling, friction, belt motion, and controllable lighting conditions. By generating fully annotated synthetic images, the digital twin serves

as a virtual testbed for both data-driven and conventional optical inspection methods, enabling systematic evaluation and training of inline quality assurance algorithms.

3. SMC Surface Analysis of production material

During the manufacturing of SMC material, there are several in-between states relevant for understanding material structure. As the SMC is generated by randomly distributing chopped fibers on a plastic carrier film, its initial randomness state gets set. Upon further compaction along the process, this state is matured further, until finally the material selection, positioning and pressing process achieves the final FOD and FVC. Among these steps in between roving chopping and finished part, the ideal position to initial FOD and FVC is the initial randomized distribution generated by the chopping and dropping of the fibers onto the plastic carrier film. Here, initial positions and orientations are set and could still be influenced. Additionally, the lack of top layer and the pre-compaction rectangular shape of rovings simplifies machine vision based approaches.

3.1. SMC material production setup

The images used as ground truth for the comparison to the digital twin generated in this work are produced on a SMC-line at the IVW. (Schmidt und Heinzmann SMC Line600) It can produce up to 600 mm of material width whilst consuming up to 23 rovings in parallel to achieve areal weights of up to 8 kg/m^2 . In addition to the common features of smc production lines such as carrier film, resin pase application, chopper, compactor and roller, the machine features an additional unwinding unit to introduce dry textiles into the process in between the chopping and application of the top layer. Due to the additional distance, an area with settled down fibers is created that used to capture the full-frame ground truth images used in this work. Here, the camera setup is placed with its additional lighting and fixtures.

The images captured for this work are taken within a production run of a usual production-run SMC (Polynt SMCarbon 80, Polynt SPA, Italy), featuring epoxy resin applied with a squeegee box gap of 1 mm. The rovings used for production are 15k tows chopped to a nominal length of 25 mm and guided onto the production width of 360 mm using baffles. The rotation speed of the chopper of 35 rpm at the carrier film speed of 2 cm/s results in an average areal weight of roughly 1 kg/m^2 . This results in a mostly

closed surface of chopped carbon fibers, leaving the carrier film only sparsely uncovered.

3.2. Camera setup to capture ground-truth data

Using the aforementioned position of the unwinding unit, the acquisition setup is placed right after the chopped fiber area. It consists of a lighting box ensuring proper light distribution within the camera frame by including LED lighting units (SBL-0130/SBL-0150, MBJ Imaging GmbH) and featuring protective, color filtering screens to allow external monitoring of the material passing through it at a lower level of lighting. The lighting box also serves as a mounting frame for the polarization camera used to capture the images. Mounted at a height of 980 mm, using a fixed focal lens (HF16HA-1B, Fujifilm Holdings Corporation, Japan) the full-frame images fully capture the 360 mm width of the production line.

The camera itself (VCXU-50MP, Baumer Electric AG, Switzerland) is an industrial usb camera for c-mount objectives featuring a monochrome sensor with a four directional on-chip polarizer (SONY IMX250MZR). The micro-wire grid polarizers are arranged directly on the chips surface as a two-by-two array similar to bayer color filter arrays as a Division-of-focal-plane sensor. It features the analyzer angles 0° , 45° , 90° and 135° . This allows simultaneous acquisition of four polarization images in one shot, capturing polarization information in real time. The resulting raw image captured by the camera can accordingly be viewed as four simultaneously captured images for the four analyzer angles, with a slight offset. From these, according to stokes formulations, all linear polarization components (S_0 , S_1 , S_2) of the original image can be calculated. For practical use with SMC, they are used to calculate two easily interpreted false-color images using the Degree of linear Polarization (DOLP) and the Angle of linear Polarization (AOLP). The DOLP image can be directly used to separate carbon fibers from unpolarized backgrounds, providing information about the FVC, whilst the AOLP directly represents the fibers' orientation and can thus be easily applied on the calculation of FOD. Among them, the AOLP can be used as the single most important image information, since it is only calculated for polarized sections of the image and thus already contains the essential information of the DOLP as well. It is accordingly the foremost focus of subjective comparison to the created synthetic image.

4. Digital Twin for SMC Production

This section describes the created set of tools, concepts and workflows for generating synthetic data that is required for developing camera based-procedures for in-line quality assurance during SMC manufacturing. It includes the generation of a large number of roving strands with constant cross section but variable length, their dropping onto the moving conveyor belt and their physical interaction with the belt, which - together with some enforced variance during the spawning of the strips - results in a realistic settlement of the strips onto the belt. However, the simulation does not explicitly include the resin, which is typically present on the conveyor belt when the fiber strips fall onto it. This is consistent with the situation during the development of camera-based inspection methods, which was performed on multiple series of images acquired directly within the chopping area. Furthermore, it is assumed that this has no significant impact on the final *distribution* of the location and orientation of the strips, at least not to an extent that defies the development of such inspection methods. The same applies to the non-inclusion of the top carrier film which is applied to the sheet before it is spooled at the end of the SMC line.

4.1. Simulation of the SMC Manufacturing Process

The newly created simulation framework consists of three main components: (i) Blender 4.0 (Blender Foundation, Amsterdam, Netherlands), an open-source 3D computer graphics software package. Blender’s integrated rigid body physics engine, based on the Bullet Physics library, is used for simulating realistic mechanical interactions between fiber strips and the conveyor belt. (ii) A self-developed Python automation script (`stripGen_Ana_automate.py`) that orchestrates scene setup, physics baking, data analysis, and rendering. (iii) An external runner script (`run_simulation.py`) that manages batch processing and parallelization across multiple simulation seeds. The runner script takes all relevant simulation parameters from a JSON configuration file (`config.json`), including the number of strips to simulate, spawn timing intervals, total simulation duration and frames for density measurement.

The simulation framework replicates a typical SMC manufacturing line consisting of a moving conveyor belt with configurable speed and friction properties, an outlet aperture positioned above the belt from which strips are spawned, and a measurement box for density quantification inside a world coordinate system whose x-axis is aligned with the belt direction of motion,

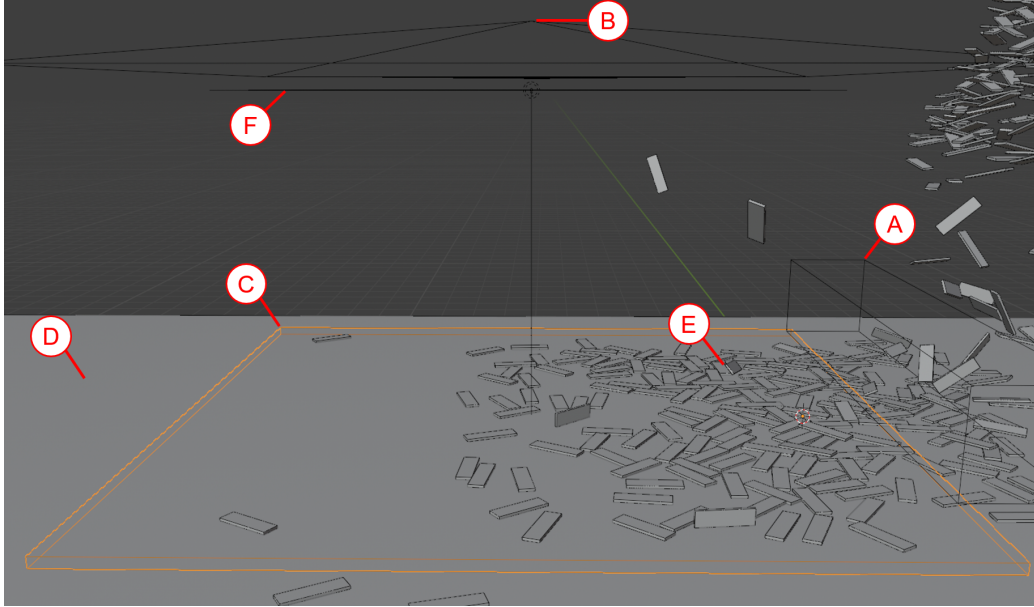


Figure 1: Simulation as displayed in Blender’s GUI. (A) shows the spawning region for the outlet, (B) indicates the camera’s origin point used for rendering, (C) marks the measurement box used for analysis and calculations, (D) represents the rigid-body conveyor-belt model, (E) shows individual fiber strips, here visibly stacked on top of one another and (F) shows the illumination area.

the y-axis parallel to the belt width, and the z-axis representing the vertical direction, see Figure 1.

The simulation framework operates in two distinct phases. In the *setup phase*, the automation script initializes the scene by clearing any existing objects, generating template geometries for strips, creating materials with appropriate visual and physical properties, and pre-computing spawn schedules to prevent initial geometric collisions. Subsequently, the physics simulation is executed across all specified frames, with the rigid body dynamics system computing trajectories and contact forces between strips and with the conveyor belt. In the *analysis phase*, executed after physics baking is complete, the framework advances to each specified measurement frame, computes both volumetric (3D) and projected (2D) density metrics within a defined measurement volume, renders polarization images for four distinct polarization angles, generates object detection annotations in YOLO (You Only Look Once) format, and persists all configuration parameters and computed metrics to an SQLite database for subsequent analysis. This separation of setup

and analysis phases, combined with deterministic random seed control, ensures that each simulation is fully reproducible while enabling efficient batch processing across parameter variations.

Individual fiber strips are represented as rectangular cuboid primitives with dimensions sampled from specified ranges. Each strip is instantiated as an independent rigid body object with assigned physical properties. Mass is computed automatically from geometry and a specified material density. Surface interaction properties include a friction coefficient, which governs the sliding resistance between strips and during fiber-to-belt contacts, and a restitution coefficient controlling energy dissipation during collisions. These parameters were calibrated to qualitatively match observed behavior of carbon fiber materials in actual SMC manufacturing (including resin).

A critical challenge in multi-object physics simulation is avoiding initial geometric overlaps that lead to unrealistic collision forces and numerical instabilities. To address this, a deterministic spawn scheduling algorithm was developed that plans spawn times and positions to minimize initial collisions.

First, for each strip, the algorithm determines the planned spawn frame based on the spawn interval parameter and the strip index. It then pre-computes geometric properties using a seeded random number generator: the strip’s width is set to 0.3 units, its thickness to 0.05 units, and its length is sampled from $[1.0, 1.1]$ units. The spawn position has a fixed x, z location, while the y -coordinate is sampled across the full width of the feeding aperture. Next, the algorithm randomizes the initial orientation to simulate natural tumbling during free fall by computing three Euler angles—yaw (around the vertical axis) from $[-5^\circ, 5^\circ]$, pitch from $[-10^\circ, 10^\circ]$, and roll from $[-10^\circ, 10^\circ]$ —which provide sufficient variance while keeping the strip predominantly horizontal. Finally, it computes the centroid vector by calculating the 3D vector to the strip’s centroid.

Next, for each strip in ascending order of planned spawn frames, the following procedure is applied until all strips are spawned. The algorithm checks the spacing relative to previously spawned strips by computing the Euclidean distance between centroids. It then computes the minimum safe distance by summing (i) the maximum of the half-extents of both strips in x, y, z , (ii) an estimate of how far the earlier strip has already fallen, and (iii) a predefined minimum clearance (default 0.8 units). If the centroid distance exceeds this minimum safe distance, the strip is spawned; otherwise, its spawn is delayed to the next frame.

The combination of minimum clearance and progressively increasing falling

distances for delayed strips effectively eliminates near-contact instabilities in the physics solver.

Each strip is spawned with zero initial velocity, subsequently acquiring motion through gravitational acceleration and aerodynamic effects implicitly captured in the physics engine’s damping parameters. A small vertical offset (0.1 length units) is applied to ensure strips are positioned above the belt surface at spawn time. The Bullet physics engine then computes strip trajectories, taking into account gravitation and contact forces and torques when strips collide with the belt surface or with other strips, thereby replicating the industrial process of dropping chopped fiber strands onto the belt.

The conveyor belt is modeled as a kinematic rigid body with predefined and constant linear velocity in the negative x -direction (default: -0.01 length units per frame). Unlike dynamic rigid bodies, kinematic bodies are not influenced by collision forces but exert forces on colliding objects. Therefore, the belt’s motion imparts forward velocity to strips through friction without being slowed. Once strips settle onto the belt surface, they acquire horizontal velocity matching the belt speed while maintaining their relative positions and orientations. The simulation continues for a specified number of frames (default: 900 frames), allowing sufficient time for all strips to spawn, drop and settle into stable configurations.

4.2. Density Calculation

Quantifying fiber distribution is required for validating simulation and for generating ground truth labels. Two complementary density metrics were implemented: a volumetric density ρ_{3D} and a projected areal density ρ_{2D} representing the coverage observed in top-down imaging.

ρ_{3D} is measured by placing a rectangular measurement volume (the so called measurement box) over the conveyor belt at a fixed location downstream from the outlet, ensuring that all strips have settled and are moving with the belt when measurements are taken. A strip is considered to be inside the measurement box if at least one of its eight corners lies within the measurement box. For each of these strips, their full strip volume is included in the total volume of strip material within the measurement box V_{strips} , resulting in a slight overestimation of the actual density when strips intersect the box boundaries. ρ_{3D} is then the ratio of V_{strips} and the volume of the measurement box V_{box} :

$$\rho_{3D} = \frac{V_{strips}}{V_{box}} \quad (1)$$

Box size and position are the same in all simulation runs. Density measurements are performed at multiple specified frames during the simulation, capturing the temporal evolution of fiber packing as strips accumulate on the belt.

Analogously, ρ_{2D} is defined as:

$$\rho_{2D} = \frac{A_{\text{shadow}}}{A_{\text{box}}} \quad (2)$$

where A_{shadow} is the total projected shadow area of all strips within the measurement box and A_{box} is the horizontal cross-sectional area of the measurement volume. When computing A_{shadow} , overlapping regions of the projected strip areas are counted only once, so ρ_{2D} measures the coverage fraction rather than the total accumulated projected area. In essence, this method of calculation results in an output identical to the use of the DoLP image to differentiate areas based on their polarizing attributes.

Density measurements are performed at multiple user-specified frames throughout the simulation, generating temporal profiles of fiber accumulation. These profiles characterize the transient behavior of the deposition process, including initial sparse coverage, gradual build-up, and eventual steady-state packing density.

4.3. Polarization Imaging Simulation

Polarization imaging is an emerging technique in industrial inspection that leverages the orientation-dependent optical properties of fibrous materials. To facilitate the development of polarization-based fiber orientation detection workflows, the proposed simulation framework includes a physically grounded polarization rendering system, thought to represent the typical hardware configuration of the used cameras, without the usual mosaicing, however.

Carbon fibers, when illuminated with unpolarized light reflect light that is generally partially linearly polarized, an effect stemming from the differences between the absorptivity of the fibers parallel and perpendicular to the fiber axis. This effect is dependant on the material composition, its structure and the wavelength distribution of the light. To measure this effect using machine vision, polarization filters are implemented to create a difference in intensity on a light-sensitive component. They again reduce the intensity of the light by filtering the polarized component, following Malus' law:

$$I_{\text{trans}}^{\text{pol}} = I_{\text{inc}}^{\text{pol}} \cos^2(\phi - \phi_a), \quad (3)$$

where ϕ is the fiber's principal axis orientation in the image plane, and ϕ_a is the orientation of the analyzer.

Real filters neither achieve full transmission in parallel orientation nor perfect extinction perpendicular to the wave orientation. Additionally, strong relationships between the wavelength and the resulting extinction ratio can be expected and are present in the camera used. Since noise and distortion as well as challenges resulting from the Dofp-Sensor are out of scope for the current work, a simplified model directly representing the DOLP with a correction factor is used in the following equation:

$$I_{\text{em}} = I_{\text{pol}} + I_{\text{unpol}} = d I_{\text{em}} + (1 - d) I_{\text{em}}, \quad (4)$$

where $d \in [0, 1]$ is the degree of linear polarization (DoLP) of the emitted light. As the unpolarized fraction is transmitted at half intensity, independent of angle:

$$I_{\text{trans}}^{\text{unpol}} = \frac{1}{2} I_{\text{inc}}^{\text{unpol}}. \quad (5)$$

The total transmitted intensity is therefore:

$$I_{\text{trans}} = I_{\text{trans}}^{\text{pol}} + I_{\text{trans}}^{\text{unpol}} \quad (6)$$

$$= d I_{\text{inc}} \cos^2(\phi - \phi_a) + \frac{1 - d}{2} I_{\text{inc}}. \quad (7)$$

Therefore, d corresponds exactly to the DoLP, as can be verified from the standard contrast-based definition:

$$DoLP = \frac{I_{\text{trans}}^{\text{max}} - I_{\text{trans}}^{\text{min}}}{I_{\text{trans}}^{\text{max}} + I_{\text{trans}}^{\text{min}}} = d, \quad (8)$$

with

$$I_{\text{trans}}^{\text{max}} = P I_{\text{inc}} + \frac{1 - P}{2} I_{\text{inc}} = \frac{1 + P}{2} I_{\text{inc}}, \quad I_{\text{trans}}^{\text{min}} = \frac{1 - P}{2} I_{\text{inc}}. \quad (9)$$

Hence, $d \in [0, 1]$ interpolates between completely unpolarized light ($d = 0$) and fully linearly polarized light ($d = 1$), enabling systematic investigation of partial polarization effects.

To simulate a division-of-focal-plane polarization camera or a rotating analyzer setup, the framework renders each measurement frame at analyzer orientations 0° , 45° , 90° and 135° . These orientations correspond to standard configurations used in Stokes parameter measurement and are sufficient to

reconstruct the full linear polarization state. For each analyzer angle, the shader’s global analyzer angle parameter is updated across all materials in the scene, and a rendering pass is executed. The orthographic camera is positioned directly above the measurement box and aligned with its principal axes, ensuring a calibrated top-down view with known geometric correspondence between image pixels and physical coordinates on the belt surface.

From the four intensity images acquired at different analyzer orientations, the pixel-wise angle of linear polarization (AoLP) is computed using standard Stokes formalism. The AoLP represents the dominant fiber orientation at each spatial location and is calculated as:

$$\text{AoLP} = \frac{1}{2} \arctan \left(\frac{I_{45^\circ} - I_{135^\circ}}{I_{0^\circ} - I_{90^\circ}} \right) \quad (10)$$

where I_{0° , I_{45° , I_{90° and I_{135° denote the intensity measured with analyzer at angle 0° , 45° , 90° and 135° respectively. To avoid numerical instabilities from division by near-zero values, a small regularization constant $\epsilon = 10^{-6}$ is added to the denominator. The resulting AoLP values span the range $[-\pi/2, +\pi/2]$, corresponding to the full range of possible linear orientations. For visualization purposes, the AoLP values are normalized to the 8-bit grayscale range $[0, 255]$ through linear scaling:

$$\text{AoLP}_{\text{norm}} = \left\lfloor \frac{\text{AoLP} + \pi/2}{\pi} \times 255 \right\rfloor \quad (11)$$

The computed AoLP images provide spatially resolved orientation maps that can serve as ground truth for training orientation-sensitive neural networks or for validating classical orientation detection algorithms. This synthetic polarization data generation capability closes a significant gap in available training datasets for the development of polarization-based industrial inspection systems. Due to the calculation of the AOLP being independent of the unpolarized components of the intensity, modeling only the polarized components within the render pipeline fulfills all the needs for generating synthetic AOLP images.

4.4. Ground Truth Data Generation

A primary motivation for developing this digital twin framework is to generate large-scale, accurately annotated training datasets for deep learning-based object detection and segmentation algorithms. The simulation environment provides perfect knowledge of object locations, geometries, and

properties, enabling automatic generation of various ground truth annotation formats.

4.4.1. YOLO Format Bounding Box Annotations

The pipeline generates object detection annotations compatible with the YOLO family of real-time detection networks. For each measurement frame, 2D bounding boxes are computed for all strip objects visible within the measurement box region. For this, all strip objects in the scene are enumerated. For each strip, its world-space vertices are obtained by applying the object’s transformation matrix to the mesh vertices. Strips are filtered based on their z-coordinate extent: a strip is included only if its vertical span overlaps with the measurement box’s vertical range.

Then, the filtered strips are projected onto the xy-plane to compute their 2D bounding boxes. For each strip, the minimum and maximum x and y coordinates are determined from its vertex set, its bounding boxes are converted to normalized world-space coordinates relative to the measurement box dimensions and the normalized coordinates are clamped to the range $[0, 1]$ to handle partial strip intersections at box boundaries.

Finally, the normalized bounding boxes are converted to YOLO format, which specifies the bounding box center coordinates and dimensions, and saved as plain text.

4.4.2. Dense Ground Truth for Segmentation

While bounding boxes provide sufficient supervision for object detection tasks, instance or semantic segmentation networks require pixel-level annotations. The simulation framework’s perfect knowledge of object geometry enables generation of precise segmentation masks.

Although not explicitly implemented in the current pipeline, the framework architecture supports extension to dense annotation formats. Potential approaches include: (1) rendering object ID passes where each strip is assigned a unique color, enabling automatic extraction of instance masks; (2) rendering semantic label maps where all strips are assigned the same label value, distinguishing foreground (strip) from background (belt); or (3) rendering depth maps and surface normal maps to provide geometric cues for downstream processing.

4.4.3. Orientation and Geometric Attributes

Beyond spatial localization, the simulation framework has access to rich per-object attributes that can serve as additional supervision signals. For each strip, the available properties include its orientation, where the strip’s 3D rotation is known in Euler angles or quaternion form and its in-plane orientation angle can be extracted from the projection onto the measurement plane; its dimensions, with exact width, length, and height recorded to support regression tasks for size estimation; and its spawn metadata, which provides temporal information such as the planned spawn frame, the actual spawn frame, and the time the strip has spent in the scene, all of which offer useful context for tracking and temporal modeling.

These attributes can be incorporated into multi-task learning frameworks or used to generate auxiliary supervision signals. For example, orientation angles can supplement bounding box annotations to train oriented object detectors (e.g., rotated bounding boxes common in aerial image analysis).

4.4.4. Annotation File Organization

The ground truth annotation system maintains a strict correspondence between rendered images and annotation files. Each frame generates four PNG images corresponding to analyzer angles 0° , 45° , 90° , 135° , a computed orientation map, and a text file containing bounding box annotations for all visible strips.

All files share a common base filename encoding the run timestamp, seed, and frame number, enabling unambiguous association and batch processing. This organizational structure is compatible with standard deep learning training pipelines and facilitates dataset versioning and management.

4.4.5. Point Annotations from Bounding Box Centroids

For density estimation tasks, bounding box annotations are converted to point annotations representing object centroids. This conversion is performed post-hoc during dataset preprocessing. For each YOLO bounding box annotation $(x_{\text{center}}, y_{\text{center}}, w, h)$ in normalized coordinates, the pixel-space centroid is computed as:

$$x_{\text{point}} = x_{\text{center}} \cdot W_{\text{img}} \tag{12}$$

$$y_{\text{point}} = y_{\text{center}} \cdot H_{\text{img}} \tag{13}$$

where W_{img} and H_{img} are the image width and height in pixels.

These point annotations are stored as CSV files with columns $\mathbf{x}$, $\mathbf{y}$, maintaining the one-to-one correspondence between image files and annotation files. This representation is particularly suited for crowd counting and density estimation networks that operate on point-based supervision rather than bounding boxes.

4.4.6. Gaussian Density Map Generation for CSRNet

For training density estimation networks such as CSRNet (Congested Scene Recognition Network), point annotations are transformed into continuous density maps through Gaussian kernel convolution. This process converts discrete object locations into a smooth spatial distribution that the network learns to predict. The *density map generation algorithm* proceeds as follows:

- **Binary Point Map Construction:** A binary map M_{binary} of the same dimensions as the input image is initialized to zero. For each annotated point (x_i, y_i) , the corresponding pixel is set to unity:

$$M_{\text{binary}}(x_i, y_i) = 1 \quad \text{for all annotated points } i \quad (14)$$

- **Gaussian Convolution:** The binary point map is convolved with a 2D Gaussian kernel G_σ with standard deviation σ :

$$D(x, y) = (M_{\text{binary}} * G_\sigma)(x, y) = \sum_{i,j} M_{\text{binary}}(i, j) \cdot G_\sigma(x - i, y - j) \quad (15)$$

where

$$G_\sigma(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \quad (16)$$

The parameter σ controls the spatial extent of influence for each object. In this work, a value of $\sigma = 4$ pixels was used as a baseline, representing a compromise between localization precision and numerical stability. The Gaussian filter is applied with constant boundary padding to avoid edge artifacts.

- **Spatial Downsampling:** Deep convolutional architectures typically produce feature maps at reduced spatial resolution due to pooling and strided convolution operations. To match the network’s output resolution, the density map is spatially downsampled by a factor corresponding to the output stride. For CSRNet, which produces outputs at 1/8

of the input resolution, the downsampling is performed through block summation:

$$D_{\text{down}}[r, c] = \sum_{i=0}^{s-1} \sum_{j=0}^{s-1} D[r \cdot s + i, c \cdot s + j] \quad (17)$$

where s is the output stride (default: 8) and $[r, c]$ index the downsampled grid.

This summation-based downsampling preserves the integral property of the density map: the sum over all pixels equals the total object count. This is a critical property for crowd counting applications, as it ensures that network predictions, when integrated over the image, yield accurate total counts.

- **Verification and Quality Control:** The validity of generated density maps is verified through conservation of mass:

$$N_{\text{objects}} \approx \sum_{x,y} D(x, y) \quad (18)$$

The summation over the density map should equal the number of annotated points within a small tolerance (typically $< 2\%$ error) introduced by boundary effects and discretization.

4.4.7. Dataset Splitting and Organization

The complete synthetic dataset is partitioned into training, validation, and optionally test subsets using stratified random sampling. Stratification is performed based on key simulation parameters (e.g., random seed, measurement frame, density level) to ensure balanced representation of different scenarios across splits. A typical split ratio follows the convention of 70% training, 15% validation, and 15% test, though this can be adjusted based on dataset size and application requirements. For each subset, images and their corresponding annotations are arranged according to the standard Ultralytics YOLO dataset directory structure. Additional annotation types—such as CSV point annotations for density-estimation tasks—follow the same split-based organization. This convention ensures compatibility with established training pipelines and allows seamless integration with tools expecting the standard YOLO dataset layout. Additionally, a YAML configuration file accompanies each dataset, specifying class labels, dataset paths, and metadata for reproducibility.

4.5. Batch Processing and Reproducibility

Large-scale synthetic data generation requires efficient batch processing infrastructure and rigorous reproducibility guarantees. The proposed framework implements several mechanisms to ensure scalability, determinism, and full traceability of simulation results.

First, all stochastic elements of the simulation—including strip dimensions, spawn positions, initial orientations, and spawn scheduling—are controlled through a seeded pseudo-random number generator (PRNG). Each simulation run is initialized with a unique integer seed value specified in the configuration file. Python’s `random` module is explicitly seeded at the beginning of each simulation with `random.seed(seed_value)`. This initialization ensures that all subsequent random number generation is deterministic and reproducible. Given identical configuration parameters and seed values, the simulation will generate pixel-perfect identical results regardless of execution time, hardware platform, or order of execution within a batch. The use of deterministic randomness not only enables exact reproduction of any dataset for validation or debugging purposes, but also supports controlled parameter studies by maintaining consistency in stochastic elements, and it facilitates collaborative research by allowing different research groups to regenerate identical training data from published configurations.

Complete reproducibility requires deterministic execution and permanent records of all input parameters. To achieve this, the framework uses a centralized JSON configuration file, which defines all simulation parameters and serves as the single source of truth. Before each batch run, a timestamped copy of this configuration is archived alongside the generated data, preserving the exact settings used even if the master file changes later. In addition, all critical parameters are stored in the database with each results entry, including simulation settings (number of strips, spawn interval, total frames), geometry (outlet position, belt speed, measurement box dimensions), random seed, frame number, and timestamps for execution and insertion. This dual storage ensures reproducibility, protects against data loss, and enables efficient querying of results by parameter values.

All simulation results are stored in a single SQLite database, providing a unified interface for analysis and dataset management. Each entry includes a unique auto-incremented `run_id`, timestamps, random seed, frame number, computed metrics (3D volumetric density, 2D areal density, object counts), and references to the corresponding rendered images and annotations. This

structure enables flexible queries, such as filtering frames by density, analyzing temporal density evolution, identifying parameter configurations with specific coverage, and generating summary statistics across many runs. The SQL interface integrates with tools like pandas, R, or MATLAB and supports automated quality control, including outlier detection and verification of annotation completeness.

4.6. Adherence to FAIR Principles

The pipeline is designed to align with the interoperability and reuseability aspects of the FAIR data principles.

All outputs are stored in open, standard formats (PNG images, CSV/TXT annotations, SQLite database, JSON configuration), requiring no proprietary software for access. Additionally, annotation formats conform to established standards (YOLO, CSV point lists) compatible with major deep learning frameworks (PyTorch, TensorFlow, Ultralytics), thus making the pipeline’s results highly interoperable.

Since all parameters, deterministic execution, and archived configurations are fully documented, exact reproduction or controlled modification of datasets by independent researchers is supported, making the pipeline also highly reusable.

5. Results and Conclusions

To evaluate the performance of the pipeline, a synthetic dataset with over 5,000 rendered images for each image component is created with annotations. Here a strong emphasis is set on the data generation methodology, in contrast to exhaustive model performance benchmarking. The focus is accordingly set on comparison to evaluable connections known from the real-world system such as spatial characteristics and preliminary validation through deep learning models.

5.1. Strip-Count Based Density Estimation

The physics simulation’s ability to produce realistic fiber accumulation dynamics is demonstrated through temporal density profiles (Fig. 2A). Density measurements were recorded at multiple frames throughout each simulation, capturing the progression from initial deposition through steady-state packing. For each parameter configuration, simulations were executed across multiple random seeds, with the shaded regions indicating the mean $\pm$ range

of observed densities. The curves exhibit characteristic behavior: an initial transient phase where strips progressively accumulate on the belt surface, followed by convergence toward a steady-state density determined by the balance between strip count and belt speed. Higher strip counts (N) produce proportionally higher equilibrium densities, while slower belt speeds (smaller $|v|$) allow more strips to accumulate within the measurement region before being transported downstream. The consistency of convergence behavior across random seeds validates both the numerical stability of the physics solver and the effectiveness of the collision-free spawn scheduling algorithm, while the observable variance quantifies the inherent stochasticity of the dropping process.

Areal weight is an important quality parameter of SMC sheets. Using identical raw materials, different areal weights can be achieved by varying the relationship of conveyor speed to fiber cutting rate and thus the amount of individual instances on a certain area of the band. For real life data, this can easily be evaluated through the 2D areal density (coverage density), which is reflected by the DOLP image component, where fibers and background can be easily separated [23]. There, a higher areal weight correlates to a higher areal density. In low areal weights, the correlation is almost linear, whilst in higher weights it can be compared to limited growth.

This correlation can be replicated by tracking the spawned strip count and comparing it to the calculated coverage, as shown in (Fig. 2A). Strip count was systematically varied over the same area to reproduce different synthetic fiber areal weights, resulting in a corresponding variation of coverage density. The controlled exploration of distribution patterns in this way represents the known correlation between fiber areal weight and coverage density. (Fig. 2B) shows representative examples spanning from sparse to dense configurations, with their corresponding metrics. They showcase the pipeline’s ability to generate a broad range of scenarios whilst automatically calculating metrics reliably. The percentage of visible background color directly represents the areal density and can thus be compared to DOLP images from real processes, although the simulation has not been tuned to the point of allowing quantitative comparisons yet.

Additionally, the simulation is controlled for consistency by simultaneous verification of the actual spawned volume of strips representing fibers against the number of generated annotations for the deep learning models, as those are essential for high training data quality. (Fig. 2C) showcases the different annotation types, which are generated simultaneously to prevent potential in-

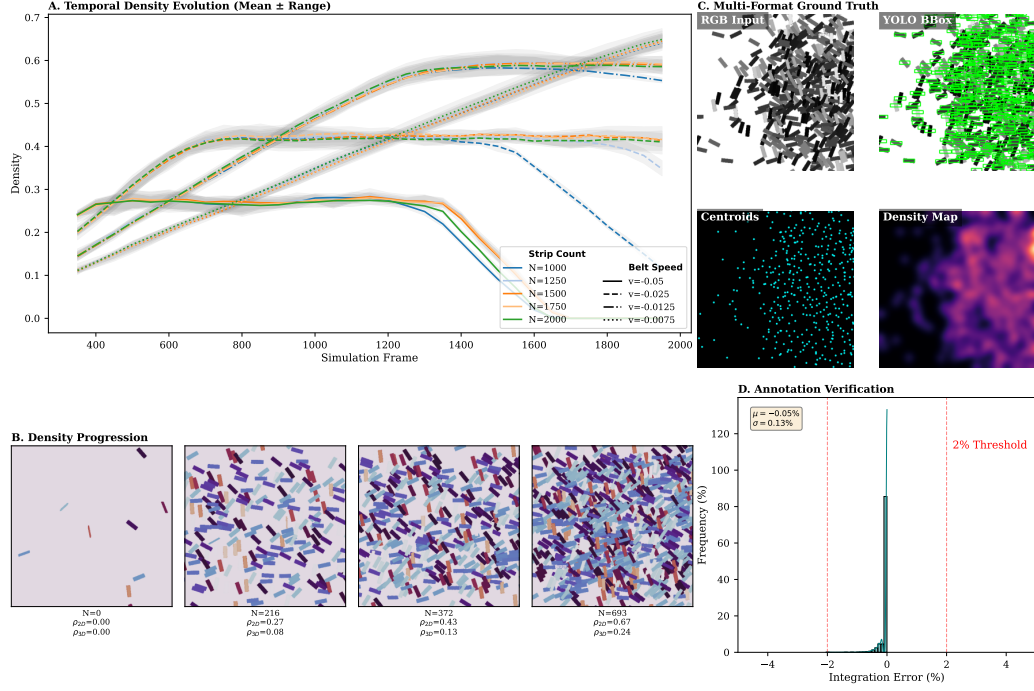


Figure 2: Systematic parameter control and multi-format ground truth generation. (A) Temporal evolution of strip density during physics simulation. Different curves represent varying strip counts (N) and belt speeds (v), with shaded regions indicating the mean $\pm$ range across multiple random seeds. The convergence to steady-state packing density after an initial transient deposition phase demonstrates reproducible simulation behavior. (B) Representative examples of rendered images with calculated metrics as used in the density calculations. (C) Input in comparison with the two generated annotation formats and the density map as used in the verification calculation. (D) Verification of the annotation as computed over all scenes, with the integral of the density matching true object count within $< 2\%$.

consistencies that could arise from sequential conversion processes. A density map is also included, providing through-thickness information more clearly than the annotation maps.

To verify annotation accuracy, a conservation of mass check is implemented by integrating the density map values across all pixels. This integral must equal the number of annotated objects, as it directly represents the number of strips spawned within the simulation. The result of these comparisons for all generated scenes is shown in (Fig. 2D). The constraint was always satisfied within a tolerance of less than 2%, with errors primarily attributable to boundary effects and spatial discretization. This verification step provides confidence that downstream density estimation models receive correctly normalized training targets.

5.2. Fiber Orientation Distribution

Fiber orientation distribution is another important parameter in in-line quality control in SMC manufacturing.

Fig. 3A shows an exemplary numeric fiber orientation distribution that was computed from 5,000 randomly chosen frames across all conducted simulations. The data were tested for uniformity using a chi-square goodness-of-fit test [28, 29]. The test yielded a test statistic of $\chi^2 = 38.2$ with $p < 0.001$, indicating that the data are consistent with a uniform distribution. This result correlates well with the quasi-isotropic distribution of commercial SMC sheets and thus demonstrates the simulation’s ability to correctly reproduce this important property of industrial SMC manufacturing [30].

The AoLP visualization examples in Fig. 3B demonstrate that orientation information is accurately captured in rendered images, making them suitable as surrogate data for actual camera images. This holds across different stages of sheet manufacturing, including the start of production, the ramp-up of areal density, and steady-state manufacturing, even though the expected areal density of 1 is not yet reached in our simulations. Consequently, these polarization-based renderings provide valuable ground truth for validating orientation detection algorithms and can serve as additional training signals for orientation-sensitive networks. The ability to generate synthetic polarization data addresses a gap in available training datasets for polarization-based industrial inspection systems.

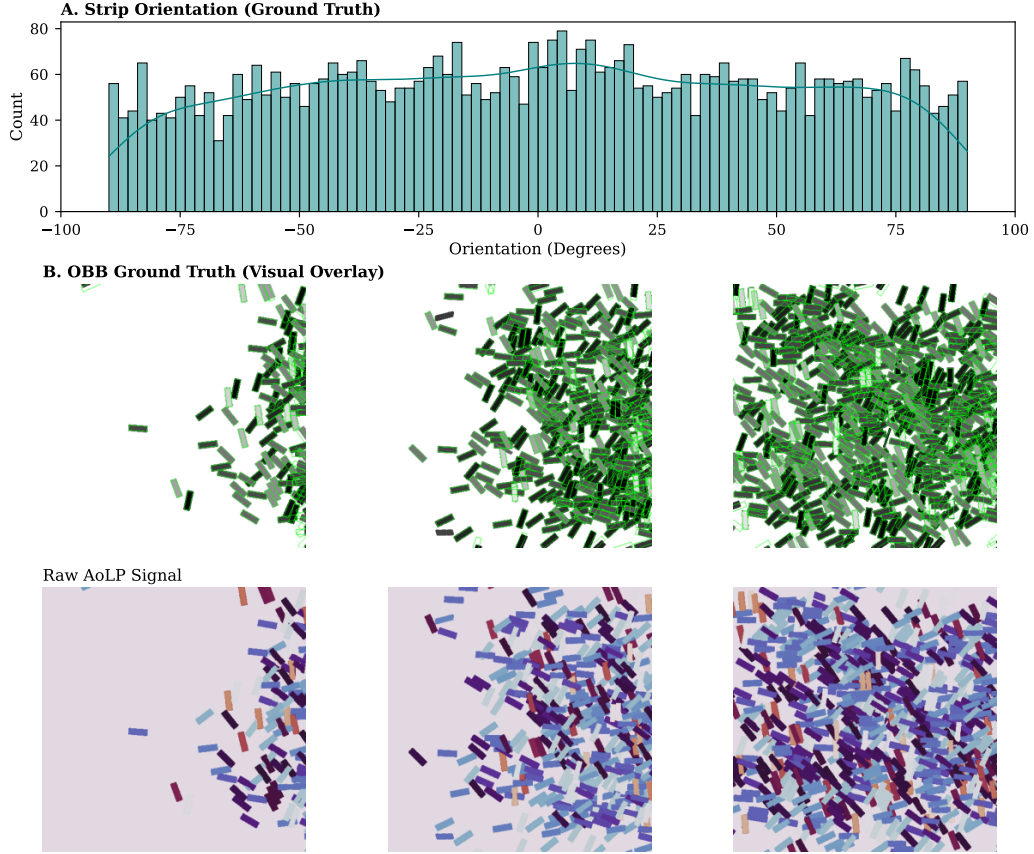


Figure 3: Fiber orientation distribution of synthetic datasets: (A) Fiber orientation distribution from 5,000 simulation frames, consistent with a uniform distribution and quasi-isotropic commercial SMC sheets. (B) AoLP renderings across manufacturing stages, capturing orientation information suitable for validating detection algorithms and training orientation-sensitive networks.

5.3. Preliminary Model Validation and Synthetic-to-Real Transfer

To assess the utility of synthetic training data for practical applications, we performed preliminary validation using representative deep learning models: YOLOv11 for object detection and CSRNet for density estimation. Both models were trained exclusively on synthetic data and evaluated on both synthetic test sets and real-world manufacturing images. The focus here is not on comprehensive model benchmarking, but rather on demonstrating that the synthetic data provides sufficient information content to enable transfer to real manufacturing scenarios.

The side-by-side comparison (Fig. 4A) illustrates systematic differences. While the synthetic and real images appear superficially similar, closer inspection reveals differences in edge characteristics, surface texture, and background appearance.

Quantitative domain gap analysis (Fig. 4B) confirms systematic distribution shifts between synthetic and real data. Intensity histograms show that synthetic images exhibit a narrower dynamic range with more uniform intensity distributions, as indicated by the different standard deviations ($\sigma_{\text{syn}} = 95.7$ vs. $\sigma_{\text{real}} = 73.9$). The real data shows a broader distribution across the intensity spectrum, attributable to fiber surface irregularities, lighting variations, and sensor noise.

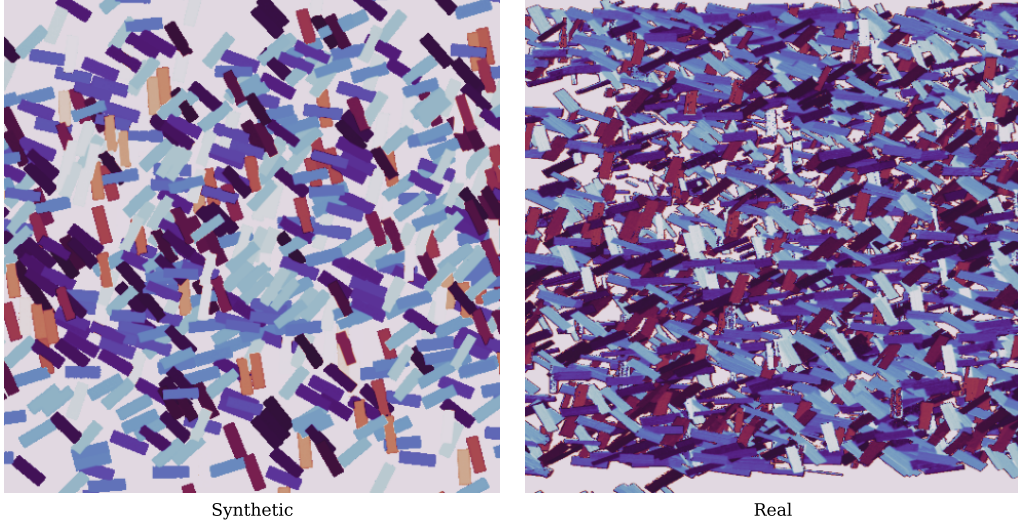
Rapid Iteration Capability. Importantly, the automated generation pipeline enables iterative refinement based on these insights within practical timeframes. New parameter configurations addressing identified gaps can be generated and integrated into the training set rapidly, compared to real-world data collection and labeling. This feedback loop transforms real-world deployment from a one-time validation step into an iterative improvement process.

Comparison to Traditional Approaches. These findings align with broader observations in synthetic-to-real transfer literature: synthetic data provides a strong initialization for perception models, but optimal performance typically requires domain adaptation techniques or inclusion of limited real labeled data. Nevertheless, the ability to bootstrap model development entirely from synthetic data represents a substantial reduction in the labeling burden.

5.4. Answers to Research Questions

This work successfully addresses all three research questions through the development and validation of a physics-based digital twin for synthetic train-

A. Visual Domain Comparison



B. Quantitative Domain Gap Analysis

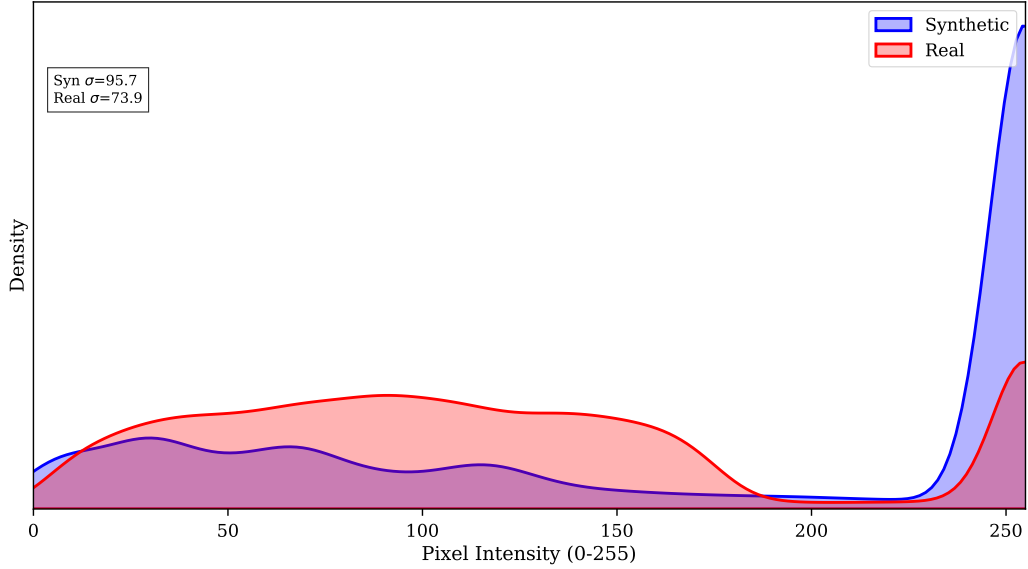


Figure 4: **Synthetic-to-real domain transfer assessment and identified challenges.** (A) Visual comparison of synthetic (left) and real (right) data at similar strip densities. While overall appearance is comparable, systematic differences in material appearance, edge sharpness, and background texture are evident upon close inspection. (B) Quantitative domain gap analysis via intensity histogram comparison. Synthetic data exhibits a narrower intensity range ($\sigma_{\text{syn}} = 95.7$) compared to real data ($\sigma_{\text{real}} = 73.9$), indicating domain shift that contributes to reduced real-world performance.

ing data generation in C-SMC strip deposition. Regarding the first research question on generating large-scale, fully annotated image datasets *in silico*, the developed simulation framework demonstrates the capability to produce over 5,000 rendered images with comprehensive annotations, capturing realistic microstructural variability through physics-based strip deposition dynamics that incorporate gravitational settling, friction, and belt motion. The framework generates multiple synchronized annotation formats—including YOLO bounding boxes, point annotations, density maps, and polarization-based renderings (AoLP and DoLP images)—while maintaining strict reproducibility through seeded random number generation and automated verification procedures that ensure annotation accuracy within 2% error. For the second research question on supporting camera-based quality assurance methods, the results demonstrate that the synthetic datasets successfully replicate key industrial characteristics: temporal density evolution profiles show realistic strip accumulation and steady-state packing behavior matching the balance between strip count and belt speed, while fiber orientation distribution analysis confirms quasi-isotropic behavior consistent with commercial SMC sheets (chi-square test, $p < 0.001$). The AoLP renderings accurately capture orientation information across different manufacturing stages, providing suitable ground truth for validating detection algorithms and training orientation-sensitive networks. Addressing the third research question on machine learning training and validation, preliminary validation using YOLOv11 for object detection and CSRNet for density estimation demonstrates that models trained exclusively on synthetic data can transfer to real manufacturing scenarios, despite an identified domain gap in intensity distributions. While systematic differences exist between synthetic and real data—particularly in edge characteristics and intensity ranges—the automated generation pipeline enables rapid iterative refinement to address these gaps, substantially reducing the labeling burden compared to manual annotation approaches while maintaining the ability to bootstrap model development entirely from synthetic data.

6. Outlook

Future work will focus on enhancing the realism and utility of synthetic data for fiber orientation analysis and anomaly detection in SMC inspection. One key area is improving the representation of fiber orientations, particularly capturing the non-uniform distributions observed in real-world scenar-

ios, which are critical for robust anomaly detection. Additionally, boundary conditions in synthetic datasets will be refined to better reflect operational realities. For example, training data should include strips partially exiting the measurement volume, in line with the sliding window inference paradigm used during deployment. This can be achieved through adjustments to camera framing or by expanding the spawn volume beyond the nominal measurement box.

Another important direction is increasing lighting diversity. The synthetic data generation pipeline will be expanded to include directional illumination at varying angles (0° – 45° from vertical), a wider range of intensities (50–200% of baseline), and both hard and soft shadows, to better capture real-world variability. Similarly, material appearance will be refined to reduce the gap with real imagery. Adjustments may include increased surface roughness, moderate variations in grayscale tone, and enhanced fiber texture detail using normal mapping to simulate surface irregularities.

Background variation will also be addressed by incorporating diverse textures and colors representative of different conveyor belt materials, such as gray, blue, or black surfaces with smooth or textured finishes, rather than the current uniform gray background. Finally, extreme density scenarios will be more heavily represented in the training data. Including cases with very high strip counts (>900) will better prepare models for worst-case real-world conditions, as current datasets emphasize moderate densities for computational efficiency.

Together, these planned improvements aim to increase the realism of synthetic training data and, in turn, enhance the reliability and performance of machine learning models for fiber orientation analysis and anomaly detection in SMC applications.

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References

- [1] J. Görthofer, N. Meyer, T. D. Pallicity, L. Schöttl, A. Trauth, M. Schemmann, M. Hohberg, P. Pinter, P. Elsner, F. Henning, A. Hrymak, T. Seelig, K. Weidenmann, L. Kärger, and T. Böhlke, “Virtual process chain of sheet molding compound: Development, validation and perspectives,” vol. 169, pp. 133–147.
- [2] T. Sabiston, K. Inal, and P. Lee-Sullivan, “Method to determine the required microstructure size to be represented by a second order fibre orientation tensor using x-ray micro computed tomography to evaluate compression moulded composites,” vol. 182, p. 107777.
- [3] M. Castro, “Numerical analysis of the compression molding process of automotive structures by means of direct fiber simulation,” pp. 655–664.
- [4] D. Anagnostou, G. Chatzigeorgiou, Y. Chemisky, and F. Meraghni, “Hierarchical micromechanical modeling of the viscoelastic behavior coupled to damage in SMC and SMC-hybrid composites,” vol. 151, pp. 8–24.
- [5] M. P. Falaschetti, F. Semprucci, L. Raimondi, and D. Serradimigni, “Experimental and numerical assessment of sheet molding compound composite crushing behavior,” vol. 34, no. 15, pp. 15281–15292.
- [6] N. Meyer, L. Schöttl, L. Bretz, A. Hrymak, and L. Kärger, “Direct bundle simulation approach for the compression molding process of sheet molding compound,” vol. 132, p. 105809.
- [7] P. Feraboli, E. Peitso, F. Deleo, T. Cleveland, and P. B. Stickler, “Characterization of prepreg-based discontinuous carbon fiber/epoxy systems,” vol. 28, no. 10, pp. 1191–1214.
- [8] M. Duhovic, A. Keilmann, D. Schommer, C. Redenbach, and K. Schladitz, “Quantitative fiber orientation analysis of carbon fiber sheet molding compounds using polarization imaging and x-ray computed tomography,” vol. 295, p. 112146.
- [9] C. Nony-Davadie, L. Peltier, Y. Chemisky, B. Surowiec, and F. Meraghni, “Mechanical characterization of anisotropy on a carbon fiber sheet

- molding compound composite under quasi-static and fatigue loading,” vol. 53, no. 11, pp. 1437–1457.
- [10] S. Garcea, Y. Wang, and P. Withers, “X-ray computed tomography of polymer composites,” vol. 156, pp. 305–319.
 - [11] G. Bardl, A. Nocke, C. Cherif, M. Pooch, M. Schulze, H. Heuer, M. Schiller, R. Kupke, and M. Klein, “Automated detection of yarn orientation in 3d-draped carbon fiber fabrics and preforms from eddy current data,” vol. 96, pp. 312–324.
 - [12] M. De Goeje and K. Wapenaar, “Non-destructive inspection of carbon fibre-reinforced plastics using eddy current methods,” vol. 23, no. 3, pp. 147–157.
 - [13] Y. Sugimoto, D. Shimamoto, Y. Hotta, and H. Niino, “Estimation of the fiber orientation distribution of carbon fiber-reinforced plastics using small-angle x-ray scattering,” vol. 9, p. 100194.
 - [14] H. Fernandes, H. Zhang, and X. Maldague, “An active infrared thermography method for fiber orientation assessment of fiber-reinforced composite materials,” vol. 72, pp. 286–292.
 - [15] C. Eberhardt and A. Clarke, “Fibre-orientation measurements in short-glass-fibre composites. part i: automated, high-angular-resolution measurement by confocal microscopy,” vol. 61, no. 10, pp. 1389–1400.
 - [16] A. Kraemer, S. Lin, D. Brabandt, T. Böhlke, and G. Lanza, “Quality control in the production process of SMC lightweight material,” vol. 17, pp. 772–777.
 - [17] S. Meister, L. Grundhöfer, J. Stüve, and R. M. Groves, “Imaging sensor data modelling and evaluation based on optical composite characteristics: Investigation of data quality for inline inspection,” vol. 116, no. 11, pp. 3965–3990.
 - [18] G. A. Atkinson, S. O’Hara Nash, and L. N. Smith, “Precision fibre angle inspection for carbon fibre composite structures using polarisation vision,” vol. 10, no. 22, p. 2765.

- [19] O. Döbrich and C. Brauner, “Machine vision system for digital twin modeling of composite structures,” vol. 10, p. 1154655.
- [20] C. Freitag, R. Weber, and T. Graf, “Polarization dependence of laser interaction with carbon fibers and CFRP,” vol. 22, no. 2, p. 1474.
- [21] D. Schommer, M. Duhovic, T. Hoffmann, J. Ernst, K. Schladitz, A. Moghiseh, F. Gortner, J. Hausmann, P. Mitschang, and K. Steiner, “Polarization imaging for surface fiber orientation measurements of carbon fiber sheet molding compounds,” vol. 37, p. 101456.
- [22] L. Chiominto, G. D’Emilia, and E. Natale, “Using light polarization to identify fiber orientation in carbon fiber components: Metrological analysis,” vol. 24, no. 17, p. 5685.
- [23] M. Duhovic, D. Schommer, M. Vu, F. Röder, F. Gortner, D. May, and K. Steiner, “3d-DIGITALIZATION OF THE FIBER ORIENTATION IN c-SMC SEMI-FINISHED PRODUCTS USING IN-SITU POLARIZATION IMAGING,” vol. 8.
- [24] N. Hütten, M. Alves Gomes, F. Hölken, K. Andricevic, R. Meyes, and T. Meisen, “Deep learning for automated visual inspection in manufacturing and maintenance: A survey of open- access papers,” vol. 7, no. 1, p. 11.
- [25] J. Teuwsen, S. K. Hohn, and T. A. Osswald, “Direct fiber simulation of a compression molded ribbed structure made of a sheet molding compound with randomly oriented carbon/epoxy prepreg strands—a comparison of predicted fiber orientations with computed tomography analyses,” vol. 4, no. 4, p. 164.
- [26] Z. Chen, T. Huang, Y. Shao, Y. Li, H. Xu, K. Avery, D. Zeng, W. Chen, and X. Su, “Multiscale finite element modeling of sheet molding compound (SMC) composite structure based on stochastic mesostructure reconstruction,” vol. 188, pp. 25–38.
- [27] J. Tremblay, A. Prakash, D. Acuna, M. Brophy, V. Jampani, C. Anil, T. To, E. Cameracci, S. Boochoon, and S. Birchfield, “Training deep networks with synthetic data: Bridging the reality gap by domain randomization,” in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pp. 1082–10828, IEEE.

- [28] K. Pearson, “On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling,” *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, vol. 50, no. 302, pp. 157–175, 1900.
- [29] J. A. Rice, *Mathematical Statistics and Data Analysis*. Belmont, CA: Duxbury Press, 3rd ed., 2007.
- [30] M. F. Ashby, “Chapter 15 - Material profiles,” in *Materials and the Environment (Second Edition)* (M. F. Ashby, ed.), pp. 459–595, Boston: Butterworth-Heinemann, second edition ed., 2013.